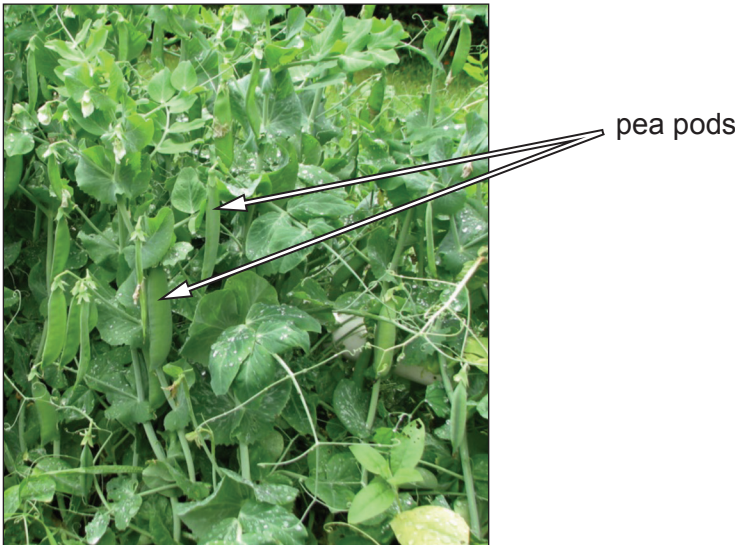


2. Image 2.1 shows pea pods growing on pea plants (*Pisum sativum*).

Image 2.1



Peas grow inside the pods.



(a) Owain wanted to see if longer pods contain more peas than shorter pods.

He:

- Picked four pods from each of the two tallest pea plants on the same day.
- Measured the length of the eight pods.
- Counted and recorded the number of peas in each of the eight pods.

The results are shown in **Table 2.2**.

Table 2.2

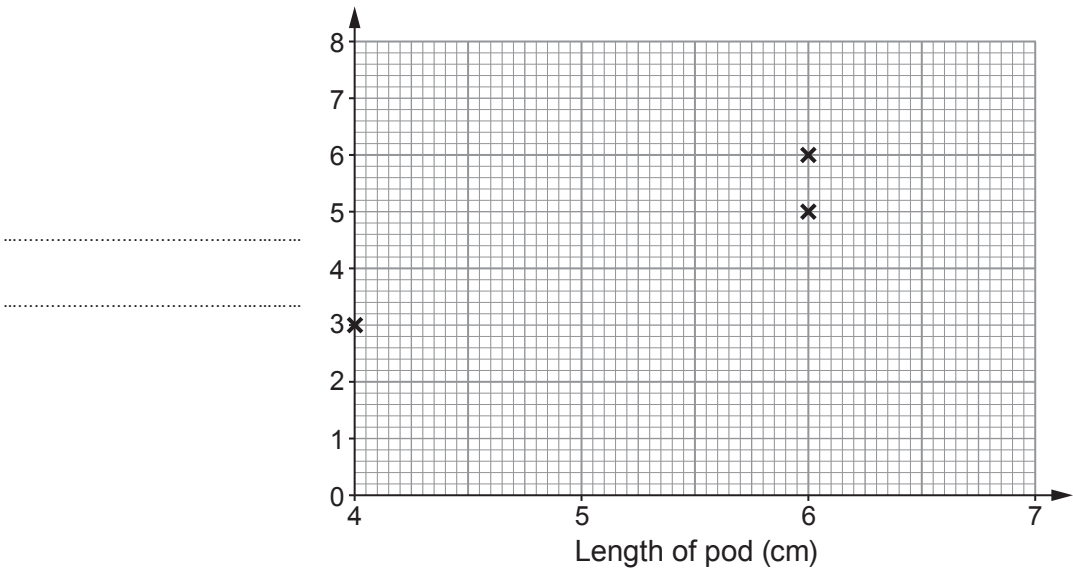
Length of pod (cm)	Number of peas in pod
6.0	6
4.0	3
6.0	5
5.5	5
4.5	4
5.0	4
6.5	7
5.5	6



- (i) Calculate the mean number of peas in the eight pods.
Space for working.
- [2]

Mean number of peas in a pod =

- (ii) Use the results in **Table 2.2** to plot a **scattergraph** by:
- I. labelling the vertical axis;
- [1]
- II. plotting the points. The first three points have been done for you (**do not join the plots**).
- [2]



- (iii) From the scattergraph:
- state the relationship between pod length and number of peas.
- [1]

.....

.....

.....

.....



(iv) Underline **two** statements from the list to show **two** ways that Owain could increase confidence in this investigation. [2]

- count the peas again
- select the plants at random
- measure pod length in millimetres
- collect more pods from each plant

(b) Owain used two different pea plants in his investigation. They grew in two different parts of his garden.
Choose the correct letter (**A**, **B** or **C**) from the list to complete the following sentence. [1]

Variation in the number of peas in the pods is caused by:

- A** genetic factors only
- B** environmental factors only
- C** both genetic and environmental factors

Answer letter =

9

